

JCPillOrganizer

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Chapter 1

Deprecated List

Member [jct.pillorganizer.device.DeviceStateWrapper.buildBinSchedule \(\)](#)

use schedule strategy directly

Class [jct.pillorganizer.model.device.DeviceSchedule](#)

This is an ugly holdover from a previous system design. Move to using `DeviceBaseScheduleStrategy` directly. Refactoring in progress.

Class [jct.pillorganizer.model.EventType](#)

should use protobuf directly

Member [jct.pillorganizer.model.EventType.getIntValue \(\)](#)

Class [jct.pillorganizer.model.user.UserType](#)

Chapter 2

Hierarchical Index

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Chapter 4

Class Documentation

4.1 jct.pillorganizer.auth.AnonAuthService Class Reference

Public Member Functions

- Authentication [getAuthentication](#) ()
- long [getUserID](#) ()

4.1.1 Detailed Description

Utilities for dealing with anonymous users.

4.1.2 Member Function Documentation

4.1.2.1 getAuthentication()

```
Authentication jct.pillorganizer.auth.AnonAuthService.getAuthentication ( )
```

Gets the authentication record, ensuring that an anonymous user is currently signed in.

Exceptions

<i>RuntimeException</i>	if no such anonymous user exists
-------------------------	----------------------------------

Returns

the authentication record of the current anonymous user

4.1.2.2 getUserID()

```
long jct.pillorganizer.auth.AnonAuthService.getUserID ( )
```

Gets the current anonymous user's ID, ensuring that an anonymous user is currently signed in.

Exceptions

<i>RuntimeException</i>	if no such anonymous user exists
-------------------------	----------------------------------

Returns

the ID of the current anonymous user

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/auth/AnonAuthService.java

4.2 jct.pillorganizer.auth.AnonymousAuthenticationProvider Class Reference

Public Member Functions

- Publisher< AuthenticationResponse > **authenticate** (HttpRequest<?> http, AuthenticationRequest<?, ?> req)

4.2.1 Detailed Description

Implements anonymous user authentication using IDs and secrets. Anonymous users have a username in the format "\$anon\$(USER ID)", have the "anon" role, and their ID is added to the "id" attribute. Note that anonymous users are not granted any other role.

See also

[jct.pillorganizer.model.user.AnonymousUser](#)

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/auth/AnonymousAuthenticationProvider.java

4.3 jct.pillorganizer.auth.AnonymousAuthenticationRequest Class Reference

Public Member Functions

- Long **getIdentity** ()
- String **getSecret** ()
- **AnonymousAuthenticationRequest** (final long id, final String secret)

4.3.1 Detailed Description

DTO for serializing anonymous authentication requests via JSON.

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/auth/AnonymousAuthenticationRequest.java

4.4 jct.pillorganizer.model.user.AnonymousUser Class Reference

Public Member Functions

- byte[] **getSecret** ()
- void **setSecret** (final byte[] secret)

Public Member Functions inherited from [jct.pillorganizer.model.user.BaseUser](#)

- Long **getId** ()
- List< [DeviceUser](#) > **getDevices** ()
- [UserRole](#) **getRole** ()
- Timestamp **getCreated** ()
- Timestamp **getUpdated** ()
- void **setId** (final Long id)
- void **setDevices** (final List< [DeviceUser](#) > devices)
- void **setRole** (final [UserRole](#) role)
- void **setCreated** (final Timestamp created)
- void **setUpdated** (final Timestamp updated)

4.4.1 Detailed Description

An `AnonymousUser` is a user who has not formally registered for an account but still needs to be associated with some sort of user for authentication and authorization purposes. This allows customers to make use of the app without having to sign up for a full account. The `secret` is designed to be stored in secure on-device storage and not shared with the user. An anonymous user can be upgraded to a full user.

See also

`jct.pillorganizer.controller.api.app.AppUserController::upgradeAnonymous(UserRegistration)`

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/model/user/AnonymousUser.java

4.5 jct.pillorganizer.repo.AnonymousUserRepository Interface Reference

The documentation for this interface was generated from the following file:

- target/delombok/jct/pillorganizer/repo/AnonymousUserRepository.java

4.6 jct.pillorganizer.controller.api.app.AppDeviceApiController Class Reference

Public Member Functions

- Set< DeviceUserDTO > **listDeviceUser** ()
- [DeviceProvision](#) **startProvisioning** (@Body [StartProvision](#) startProvision, @Header("X-Local-TZ") @Nullable String timezone)
- [HttpServletResponse](#) <?> **checkProvisionStatus** (@QueryValue long id, @Body [VerifyProvision](#) vp)
- [Device](#) **device** (@QueryValue long id)
- void **reload** (@QueryValue long id)
- DeviceUserDTO **setDeviceSettings** (@[DeviceABAC](#)(idType=DeviceABACIDType.DEVICE) @PathVariable("id") long deviceId, @Body UpdateDeviceUserSettings dto)
- [HttpServletResponse](#) <?> **sync** (@[DeviceABAC](#)(idType=DeviceABACIDType.DEVICE) @PathVariable("id") long deviceId, @Body String body) throws InvalidProtocolBufferException
- DeviceStateDTO **consolidatedState** (@[DeviceABAC](#)(idType=DeviceABACIDType.DEVICE) @PathVariable long id, @QueryValue("date") String dateString) throws ParseException

4.6.1 Detailed Description

API endpoints for the app to configure and view device information and state.

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/controller/api/app/AppDeviceApiController.java

4.7 jct.pillorganizer.Application Class Reference

Static Public Member Functions

- static void **main** (String[] args)

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/Application.java

4.8 jct.pillorganizer.controller.api.app.AppMedicationController Class Reference

Public Member Functions

- Flux< [ScheduledMedication](#) > **medications** (@PathVariable("id") long deviceId)
- [ScheduledMedication](#) **saveMedication** (@PathVariable("id") long deviceId, @Body SaveMedicationDTO model)
- void **deleteMedication** (@PathVariable("id") long deviceId, @PathVariable("med_id") long medicationID)
- Mono< [ScheduledMedication](#) > **get** (@PathVariable("id") long deviceId, @PathVariable("med_id") long medicationID)

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/controller/api/app/AppMedicationController.java

4.9 jct.pillorganizer.controller.api.app.AppScheduleController Class Reference

Public Member Functions

- SimpleScheduleDTO **dispenseTimes** (@PathVariable("id") long deviceId)
- SimpleScheduleDTO **updateDispenseTime** (@PathVariable("id") long deviceId, @Body SimpleSchedule← DTO dto)

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/controller/api/app/AppScheduleController.java

4.10 jct.pillorganizer.controller.api.app.AppUserController Class Reference

Public Member Functions

- Mono< [AnonymousUser](#) > **anonymousRegistration** () throws NoSuchAlgorithmException
- Mono< UserInfoDTO > **authenticationStatus** ()
- Mono< [User](#) > **upgradeAnonymous** (@Body @Valid [UserRegistration](#) registration)
- Mono< [User](#) > **register** (@Body @Valid [UserRegistration](#) registration)

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/controller/api/app/AppUserController.java

4.11 jct.pillorganizer.model.user.Authenticatable Interface Reference

Public Member Functions

- long **getId** ()
- [UserType](#) **getUserType** ()

4.11.1 Detailed Description

Represents something that can act like an authenticated user in the system. This application has a few different classes of users, including devices, users, and anonymous users. Classes that implement this interface signify that the class can be used to authenticate with the system. TODO: refactor this concept to better integrate with BaseUser and authentication subsystem

The documentation for this interface was generated from the following file:

- target/delombok/jct/pillorganizer/model/user/Authenticatable.java

4.12 jct.pillorganizer.auth.AuthService Class Reference

Public Member Functions

- boolean [checkPassword](#) ([User](#) user, char[] password)
- byte[] [hashPassword](#) (char[] plaintext)
- char[] [toCharArray](#) (String plaintextPassword)
- long [getUserID](#) ()
- Mono< [Device](#) > [accessDeviceAsync](#) (long deviceID)
- [Device](#) [accessDevice](#) (long deviceID)
- boolean [isAdmin](#) ()

4.12.1 Detailed Description

Utility functions for dealing with authentication and authorization. Plaintext passwords are stored as a character array for security, since it helps prevent accidental leakage through logging.

4.12.2 Member Function Documentation

4.12.2.1 [accessDevice\(\)](#)

```
Device jct.pillorganizer.auth.AuthService.accessDevice (  
    long deviceID )
```

Attempt to access a device using the currently logged-in user's credentials.

Parameters

<i>deviceID</i>	device ID to access
-----------------	---------------------

Returns

the Device object if the user has access to the specified device

Exceptions

<i>RuntimeException</i>	if no device with the specified ID exists
<i>AuthenticationException</i>	if the user doesn't have access to the specified device

4.12.2.2 [accessDeviceAsync\(\)](#)

```
Mono< Device > jct.pillorganizer.auth.AuthService.accessDeviceAsync (  
    long deviceID )
```

Attempt to asynchronously access a device using the currently logged-in user's credentials.

Parameters

<i>deviceID</i>	device ID to access
-----------------	---------------------

Returns

a Mono wrapping a future device

4.12.2.3 checkPassword()

```
boolean jct.pillorganizer.auth.AuthService.checkPassword (
    User user,
    char[] password )
```

Check if the specified password matches the user's password.

Parameters

<i>user</i>	the user to check the password for
<i>password</i>	a character array of the password to check

Returns

true if the password matches, false otherwise

4.12.2.4 getUserID()

```
long jct.pillorganizer.auth.AuthService.getUserID ( )
```

Gets the currently logged-in user ID. Note that if the user type is Device, this will return a device ID and not a user ID.

Exceptions

<i>AuthenticationException</i>	if no user is currently signed in
--------------------------------	-----------------------------------

Returns

a user ID or device ID of the currently logged-in user

4.12.2.5 hashPassword()

```
byte[] jct.pillorganizer.auth.AuthService.hashPassword (
    char[] plaintext )
```

Hashes a plaintext password into a secure hash format.

Parameters

<i>plaintext</i>	the plaintext password to hash, as a character array.
------------------	---

Returns

the hashed password as a byte array

4.12.2.6 isAdmin()

```
boolean jct.pillorganizer.auth.AuthService.isAdmin ( )
```

Check if the currently logged-in user has the administrator role.

Returns

true if the logged-in user is an administrator, false otherwise

4.12.2.7 toCharArray()

```
char[] jct.pillorganizer.auth.AuthService.toCharArray (
    String plaintextPassword )
```

Converts a plaintext password in string form to a character array.

Parameters

<i>plaintextPassword</i>	a plaintext password in string form
--------------------------	-------------------------------------

Returns

the plaintext password as a character array

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/auth/AuthService.java

4.13 jct.pillorganizer.model.user.BaseUser Class Reference**Public Member Functions**

- Long **getId** ()
- List< [DeviceUser](#) > **getDevices** ()
- [UserRole](#) **getRole** ()
- Timestamp **getCreated** ()
- Timestamp **getUpdated** ()
- void **setId** (final Long id)
- void **setDevices** (final List< [DeviceUser](#) > devices)
- void **setRole** (final [UserRole](#) role)
- void **setCreated** (final Timestamp created)
- void **setUpdated** (final Timestamp updated)

4.13.1 Detailed Description

A generic human user account that can log in to the system. A `BaseUser` has a role, which indicates global permissions e.g., administrator. Roles may be indicated in the app's JWT tokens, so use care. A `BaseUser` can be associated with one or more devices, which is the key separation between a user and an `Authenticatable`.

See also

`Authenticatable`

The documentation for this class was generated from the following file:

- `target/delombok/jct/pillorganizer/model/user/BaseUser.java`

4.14 jct.pillorganizer.service.BinService Class Reference

Public Member Functions

- `int getBinID (DeviceClass deviceClass, DayOfWeek dayOfWeek, char period)`

4.14.1 Detailed Description

Service for dealing with bins and bin IDs on devices.

4.14.2 Member Function Documentation

4.14.2.1 getBinID()

```
int jct.pillorganizer.service.BinService.getBinID (
    DeviceClass deviceClass,
    DayOfWeek dayOfWeek,
    char period )
```

Finds the bin ID for the given specification. For example, this can convert (MONDAY, 'A') into bin ID 1.

Parameters

<i>deviceClass</i>	type of device
<i>dayOfWeek</i>	day of the week
<i>period</i>	period of the day, 'A' for AM and 'P' for PM

Returns

best bin ID for the given parameters

The documentation for this class was generated from the following file:

- `target/delombok/jct/pillorganizer/service/BinService.java`

4.15 jct.pillorganizer.model.device.BinStatus Enum Reference

Public Member Functions

- **BinStatus** (int val)
- int [getIntValue](#) ()

Static Public Member Functions

- static [BinStatus fromProtobuf](#) (Pill.BinStatus st)

Public Attributes

- **DISABLED** =(0)
- **TAKEN** =(1)
- **MISSED** =(2)
- **PENDING** =(3)
- **TAKE_NOW** =(4)

4.15.1 Member Function Documentation

4.15.1.1 fromProtobuf()

```
static BinStatus jct.pillorganizer.model.device.BinStatus.fromProtobuf (
    Pill.BinStatus st ) [static]
```

Converts a protobuf `BinStatus` into this Java enum.

Parameters

<code>st</code>	a protobuf <code>BinStatus</code>
-----------------	-----------------------------------

Returns

java enum

4.15.1.2 getIntValue()

```
int jct.pillorganizer.model.device.BinStatus.getIntValue ( )
```

Converts the status into an integer value compatible with how the firmware stores bin status.

Returns

the integer value of the bin status

The documentation for this enum was generated from the following file:

- target/delombok/jct/pillorganizer/model/device/BinStatus.java

4.16 jct.pillorganizer.serde.CharEncodeSerde Class Reference

Public Member Functions

- Character **deserialize** (Decoder decoder, DecoderContext context, Argument<? super Character > type) throws IOException
- void **serialize** (Encoder encoder, EncoderContext context, Argument<? extends Character > type, Character value) throws IOException

4.16.1 Detailed Description

Serializes a char into a simple string with a single character.

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/serde/CharEncodeSerde.java

4.17 jct.pillorganizer.model.device.DayOfWeek Enum Reference

Public Member Functions

- **DayOfWeek** (int val)
- int [getIntValue](#) ()

Static Public Member Functions

- static [DayOfWeek fromProtoBuf](#) (Pill.BinSchedule.DayOfWeek eventType)
- static [DayOfWeek fromOrdinal](#) (int ord)

Public Attributes

- **DISABLED** =(1000)
- **MONDAY** =(0)
- **TUESDAY** =(1)
- **WEDNESDAY** =(2)
- **THURSDAY** =(3)
- **FRIDAY** =(4)
- **SATURDAY** =(5)
- **SUNDAY** =(6)

4.17.1 Detailed Description

Each day of the week, and an application-specific "DISABLED" day, indicating that the day is to be disregarded.

4.17.2 Member Function Documentation

4.17.2.1 fromOrdinal()

```
static DayOfWeek jct.pillorganizer.model.device.DayOfWeek.fromOrdinal (
    int ord ) [static]
```

Converts a Monday-base DayOfWeek ordinal into an enum.

Parameters

<i>ord</i>	an integer DayOfWeek value
------------	----------------------------

Returns

java enum

4.17.2.2 fromProtobuf()

```
static DayOfWeek jct.pillorganizer.model.device.DayOfWeek.fromProtobuf (
    Pill.BinSchedule.DayOfWeek eventType ) [static]
```

Converts a DayOfWeek protobuf enum to a Java enum.

Parameters

<i>eventType</i>	protobuf DayOfWeek
------------------	--------------------

Returns

Java DayOfWeek

4.17.2.3 getIntValue()

```
int jct.pillorganizer.model.device.DayOfWeek.getIntValue ( )
```

Converts the DayOfWeek into an integer form based on Monday as day zero.

Returns

integer value of the DayOfWeek where Monday is the start of the week

The documentation for this enum was generated from the following file:

- target/delombok/jct/pillorganizer/model/device/DayOfWeek.java

4.18 jct.pillorganizer.model.device.Device Class Reference**Public Member Functions**

- long *getId* ()
- UserType *getUserType* ()
- ZoneOffset *getTimeZone* ()
- DeviceClass *getDeviceClass* ()
- long *getSerialNo* ()

- Set< [DeviceEvent](#) > **getEvents** ()
 - List< [DeviceState](#) > **getState** ()
 - [DeviceProvision](#) **getCurrentProvision** ()
 - [DeviceBaseScheduleStrategy](#) **getScheduleStrategy** ()
 - List< [DeviceProvision](#) > **getProvisions** ()
 - Long **getStateHash** ()
 - String **getEngrReq** ()
 - Long **getVersion** ()
 - long **getEventCounter** ()
 - String **getCustomName** ()
 - List< [DeviceUser](#) > **getUsers** ()
 - Timestamp **getLastSync** ()
 - Integer **getIpv4** ()
 - byte[] **getIpv6** ()
 - String **getBaseTZ** ()
 - void **setId** (final Long id)
 - void **setDeviceClass** (final [DeviceClass](#) deviceClass)
 - void **setSerialNo** (final long serialNo)
 - void **setEvents** (final Set< [DeviceEvent](#) > events)
 - void **setState** (final List< [DeviceState](#) > state)
 - void **setCurrentProvision** (final [DeviceProvision](#) currentProvision)
 - void **setScheduleStrategy** (final [DeviceBaseScheduleStrategy](#) scheduleStrategy)
 - void **setProvisions** (final List< [DeviceProvision](#) > provisions)
 - void **setStateHash** (final Long stateHash)
 - void **setEngrReq** (final String engrReq)
 - void **setVersion** (final Long version)
 - void **setEventCounter** (final long eventCounter)
 - void **setCustomName** (final String customName)
 - void **setUsers** (final List< [DeviceUser](#) > users)
 - void **setLastSync** (final Timestamp lastSync)
 - void **setIpv4** (final Integer ipv4)
 - void **setIpv6** (final byte[] ipv6)
 - void **setBaseTZ** (final String baseTZ)
-
- long **getId** ()
 - [UserType](#) **getUserType** ()

4.18.1 Detailed Description

A pill organizer device.

4.18.2 Member Function Documentation

4.18.2.1 getId()

```
long jct.pillorganizer.model.device.Device.getId ( )
```

Implements [jct.pillorganizer.model.user.Authenticatable](#).

4.18.2.2 `getUserType()`

`UserType` `jct.pillorganizer.model.device.Device.getUserType ( )`

Implements `jct.pillorganizer.model.user.Authenticatable`.

The documentation for this class was generated from the following file:

- `target/delombok/jct/pillorganizer/model/device/Device.java`

4.19 `jct.pillorganizer.auth.DeviceABAC` Interface Reference

Public Member Functions

- `DeviceABACIDType` `idType ()` default `DeviceABACIDType.NONE`

The documentation for this interface was generated from the following file:

- `target/delombok/jct/pillorganizer/auth/DeviceABAC.java`

4.20 `jct.pillorganizer.auth.DeviceABACIDType` Enum Reference

Public Attributes

- `NONE`
- `DEVICE`

The documentation for this enum was generated from the following file:

- `target/delombok/jct/pillorganizer/auth/DeviceABACIDType.java`

4.21 `jct.pillorganizer.auth.DeviceABACInterceptor` Class Reference

Public Member Functions

- Object `intercept` (`MethodInvocationContext`< `Object`, `Object` > `context`)

The documentation for this class was generated from the following file:

- `target/delombok/jct/pillorganizer/auth/DeviceABACInterceptor.java`

4.22 jct.pillorganizer.controller.api.device.DeviceAPIv12Controller Class Reference

Public Member Functions

- `HttpResponse<?> sync (@Body byte[] body)` throws `InvalidProtocolBufferException`
- `HttpResponse<?> provision (@Body byte[] body)` throws `InvalidProtocolBufferException`

4.22.1 Detailed Description

Endpoints designed to be called only by pill organizers.

The documentation for this class was generated from the following file:

- `target/delombok/jct/pillorganizer/controller/api/device/DeviceAPIv12Controller.java`

4.23 jct.pillorganizer.auth.DeviceAuthenticationProvider Class Reference

Public Member Functions

- `Publisher< AuthenticationResponse > authenticate (HttpRequest<?> httpReq, AuthenticationRequest<?, ?> authReq)`

4.23.1 Detailed Description

Implements device authentication using their serial number and OOB provisioning key (shared secret). Device users are given usernames in the format "\$dev#(SERIAL NUMBER)" and are given the role "device" with the attributes "sn" for the serial number and "id" for the device ID.

The documentation for this class was generated from the following file:

- `target/delombok/jct/pillorganizer/auth/DeviceAuthenticationProvider.java`

4.24 jct.pillorganizer.auth.DeviceAuthenticationRequest Class Reference

Public Member Functions

- `Long getIdentity ()`
- `Pill.AuthorizeRequest getSecret ()`
- `DeviceAuthenticationRequest (final long serialNo, final Pill.AuthorizeRequest req)`

4.24.1 Detailed Description

Wraps a protobuf `AuthorizeRequest` for use in the Micronaut Security framework.

The documentation for this class was generated from the following file:

- `target/delombok/jct/pillorganizer/auth/DeviceAuthenticationRequest.java`

4.25 jct.pillorganizer.auth.DeviceAuthService Class Reference

Public Member Functions

- Authentication [getAuthentication](#) ()
- long [getSerialNo](#) ()
- long [getDeviceID](#) ()
- [Device](#) [getDevice](#) ()

4.25.1 Detailed Description

Utilities for dealing with devices.

4.25.2 Member Function Documentation

4.25.2.1 getAuthentication()

```
Authentication jct.pillorganizer.auth.DeviceAuthService.getAuthentication ( )
```

Gets the authentication record of the currently logged-in device, ensuring that a device is logged in.

Exceptions

<i>RuntimeException</i>	if no device is logged in
-------------------------	---------------------------

Returns

the device's authentication record

4.25.2.2 getDevice()

```
Device jct.pillorganizer.auth.DeviceAuthService.getDevice ( )
```

Gets the currently logged-in device's `Device` object

Exceptions

<i>RuntimeException</i>	if no device authentication exists
-------------------------	------------------------------------

Returns

the current device's domain object

4.25.2.3 getDeviceID()

```
long jct.pillorganizer.auth.DeviceAuthService.getDeviceID ( )
```

Gets the currently logged-in device's ID

Exceptions

<i>RuntimeException</i>	if no device authentication exists
-------------------------	------------------------------------

Returns

the current device's ID

4.25.2.4 getSerialNo()

```
long jct.pillorganizer.auth.DeviceAuthService.getSerialNo ( )
```

Gets the currently logged-in device's serial number

Exceptions

<i>RuntimeException</i>	if no device authentication exists
-------------------------	------------------------------------

Returns

the current device's serial number

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/auth/DeviceAuthService.java

4.26 jct.pillorganizer.model.device.schedule.DeviceBaseDispenseTime Class Reference

Public Member Functions

- Long **getId** ()

- [DeviceBaseScheduleStrategy](#) **getSchedule** ()
- List< [MedicationDispenseTime](#) > **getMedications** ()
- void **setId** (final Long id)
- void **setSchedule** (final [DeviceBaseScheduleStrategy](#) schedule)
- void **setMedications** (final List< [MedicationDispenseTime](#) > medications)

4.26.1 Detailed Description

A single medication dispense time, for a particular device's schedule. Medications are attached to a dispense time to indicate they should be taken then.

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/model/device/schedule/DeviceBaseDispenseTime.java

4.27 [jct.pillorganizer.model.device.schedule.DeviceBaseScheduleStrategy](#) Class Reference

Public Member Functions

- abstract Object **buildDTO** ()
- Long **getId** ()
- [Device](#) **getDevice** ()
- Set< [DeviceBaseDispenseTime](#) > **getTimes** ()
- void **setId** (final Long id)
- void **setDevice** (final [Device](#) device)
- void **setTimes** (final Set< [DeviceBaseDispenseTime](#) > times)

4.27.1 Detailed Description

An abstract device schedule. A device schedule has zero or more "dispense times", indicating when medication should be dispensed on a device. This class is abstract to possibly support different scheduling styles, i.e., once a day or every other day.

4.27.2 Member Function Documentation

4.27.2.1 buildDTO()

```
abstract Object jct.pillorganizer.model.device.schedule.DeviceBaseScheduleStrategy.buildDTO (
) [abstract]
```

Converts the schedule into a data transfer object for JSON serialization.

See also

[DeviceSimpleScheduleStrategy::buildDTO\(\)](#)

Returns

a DTO of this schedule

Reimplemented in [jct.pillorganizer.model.device.schedule.DeviceSimpleScheduleStrategy](#).

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/model/device/schedule/DeviceBaseScheduleStrategy.java

4.28 jct.pillorganizer.model.device.DeviceBinId Class Reference

Public Member Functions

- long **getDeviceID** ()
- int **getBinID** ()
- void **setDeviceID** (final long deviceID)
- void **setBinID** (final int binID)
- boolean **equals** (final java.lang.Object o)
- int **hashCode** ()
- java.lang.String **toString** ()
- **DeviceBinId** (final long deviceID, final int binID)

Protected Member Functions

- boolean **canEqual** (final java.lang.Object other)

4.28.1 Detailed Description

An identifier referencing a specific bin on a specific pill organizer.

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/model/device/DeviceBinId.java

4.29 jct.pillorganizer.model.device.DeviceClass Enum Reference

Public Member Functions

- **DeviceClass** (int binCount)
- int **getBinCount** ()

Public Attributes

- **v1_7x2** =(14)

4.29.1 Detailed Description

Describes the "type" of pill organizer a device is. There is presently only a single pill organizer model. This enum is to possibly support multiple models in the future, possibly with wildly different bin configurations.

The documentation for this enum was generated from the following file:

- target/delombok/jct/pillorganizer/model/device/DeviceClass.java

4.30 jct.pillorganizer.model.device.DeviceEvent Class Reference

Public Member Functions

- Long **getId** ()
- [Device](#) **getDevice** ()
- Instant **getTs** ()
- [EventType](#) **getEventType** ()
- int **getBin** ()
- void **setId** (final Long id)
- void **setDevice** (final [Device](#) device)
- void **setTs** (final Instant ts)
- void **setEventType** (final [EventType](#) eventType)
- void **setBin** (final int bin)

4.30.1 Detailed Description

A single event recorded by a pill organizer device.

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/model/device/DeviceEvent.java

4.31 jct.pillorganizer.repo.DeviceEventRepository Interface Reference

Public Member Functions

- Collection< [DeviceEvent](#) > **findByDeviceID** (long id)

The documentation for this interface was generated from the following file:

- target/delombok/jct/pillorganizer/repo/DeviceEventRepository.java

4.32 jct.pillorganizer.model.device.DeviceProvision Class Reference

Public Member Functions

- Long **getId** ()
- [Device](#) **getDevice** ()
- boolean **isActive** ()
- byte[] **getOobKey** ()
- String **getBssid** ()
- String **getSsid** ()
- Long **getVersion** ()
- [BaseUser](#) **getUser** ()
- long **getUserID** ()
- Timestamp **getCreated** ()

- Timestamp **getUpdated** ()
- String **getTimezone** ()
- void **setId** (final Long id)
- void **setDevice** (final [Device](#) device)
- void **setActive** (final boolean active)
- void **setOobKey** (final byte[] oobKey)
- void **setBssid** (final String bssid)
- void **setSsid** (final String ssid)
- void **setVersion** (final Long version)
- void **setUser** (final [BaseUser](#) user)
- void **setUserID** (final long userID)
- void **setCreated** (final Timestamp created)
- void **setUpdated** (final Timestamp updated)
- void **setTimezone** (final String timezone)

4.32.1 Detailed Description

A record of a particular attempt at provisioning a pill organizer device, regardless of it was successful or not. This is essentially a "session" record of a provision attempt. A device can potentially have many different provision records.

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/model/device/DeviceProvision.java

4.33 jct.pillorganizer.repo.DeviceProvisionRepository Interface Reference

Public Member Functions

- Optional< [DeviceProvision](#) > **findByIdAndDevice_SerialNo** (long id, long sn)
- Optional< [DeviceProvision](#) > **findBySerialNo** (long sn)
- void **updateActiveByDeviceAndActive** ([Device](#) device, boolean active, boolean newActive)
- Optional< [DeviceProvision](#) > **findActiveBySerialNo** (long sn)
- void **update** (@Id Long id, @Version Long version, String bssid, String ssid)

The documentation for this interface was generated from the following file:

- target/delombok/jct/pillorganizer/repo/DeviceProvisionRepository.java

4.34 jct.pillorganizer.service.DeviceProvisionService Class Reference

Public Member Functions

- [DeviceProvision](#) **startProvisioning** (long serialNo, [DeviceClass](#) deviceClass, String tzLocation)
- [Device](#) **checkProvisioning** (long provID, long sn, String ssid)
- [Pill.SyncResponse](#) **completeProvisioning** ([Pill.DeviceProvisionRequest](#) req)

4.34.1 Detailed Description

Business logic for dealing with device provisioning.

4.34.2 Member Function Documentation

4.34.2.1 checkProvisioning()

```
Device jct.pillorganizer.service.DeviceProvisionService.checkProvisioning (
    long provID,
    long sn,
    String ssid )
```

Check if a device's provisioning process has completed. The parameters must exactly match what was provided during the provisioning process otherwise this method will fail.

Parameters

<i>provID</i>	the ID of the provision record to check
<i>sn</i>	serial number of the device
<i>ssid</i>	SSID of the WiFi network the device was provisioned for

Returns

the Device record, if provisioning is finished & successful

Exceptions

<i>org.zalando.problem.ThrowableProblem</i>	if provisioning has not finished, was not successful, or the parameters were invalid.
---	---

4.34.2.2 completeProvisioning()

```
Pill.SyncResponse jct.pillorganizer.service.DeviceProvisionService.completeProvisioning (
    Pill.DeviceProvisionRequest req )
```

Finish provisioning, indicating that the device has successfully connected to WiFi (thus, this is called from the device itself).

Parameters

<i>req</i>	protobuf structure with information about provisioning
------------	--

Returns

device sync response

4.34.2.3 startProvisioning()

```
DeviceProvision jct.pillorganizer.service.DeviceProvisionService.startProvisioning (
    long serialNo,
    DeviceClass deviceClass,
    String tzLocation )
```

Initiates the provisioning flow. If no Device record exists with the specified serial number, a new Device is created. The serial number must be accurate otherwise provisioning will eventually fail. The timezone is written into the Device's timezone record and should be set to the user's timezone. A successful call to this method will invalidate all previous provision attempts (but will keep an existing device provisioned until completion). Provisioning is initiated by the app.

Parameters

<i>serialNo</i>	the serial number of the device to provision
<i>deviceClass</i>	the type of device (device class) of the device to provision
<i>tzLocation</i>	timezone location string (e.g., America/Detroit) to base the device's time on

Returns

initialized provisioning record

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/service/DeviceProvisionService.java

4.35 jct.pillorganizer.repo.DeviceRepository Interface Reference

Public Member Functions

- Optional< Device > **findBySerialNo** (long serialNo)
- List< Device > **findAllProvisioned** ()
- List< Device > **findByUserID** (long userID)
- List< Device > **findAll** ()
- void **update** (@Id Long id, @Version Long version, long stateHash, long eventCounter)
- void **update** (@Id Long id, @Version Long version, DeviceProvision currentProvision)
- void **updateLastSyncAndIpv4AndIpv6** (@Id Long id, @Version Long version, Timestamp lastSync, @Nullable Integer ipv4, @Nullable byte[] ipv6)
- void **update** (@Id Long id, String customName)
- void **updateBaseTZById** (@Id Long id, String baseTZ)
- Optional< Device > **findBySerialNoAndCurrentProvisionOobKey** (long sn, byte[] oob)
- Optional< Device > **findByCurrentProvisionOobKey** (byte[] oob)

The documentation for this interface was generated from the following file:

- target/delombok/jct/pillorganizer/repo/DeviceRepository.java

4.36 jct.pillorganizer.model.device.DeviceSchedule Class Reference

Public Member Functions

- int **getBinID** ()
- [DeviceBinId](#) **getId** ()
- [Device](#) **getDevice** ()
- [DayOfWeek](#) **getDayOfWeek** ()
- int **getSecondsFrom00** ()
- [DeviceBaseDispenseTime](#) **getDispenseTime** ()
- Long **getVersion** ()
- void **setId** (final [DeviceBinId](#) id)
- void **setDevice** (final [Device](#) device)
- void **setDayOfWeek** (final [DayOfWeek](#) dayOfWeek)
- void **setSecondsFrom00** (final int secondsFrom00)
- void **setDispenseTime** (final [DeviceBaseDispenseTime](#) dispenseTime)
- void **setVersion** (final Long version)

4.36.1 Detailed Description

Represents the schedule a single bin on a particular pill organizer should follow, such as the day of the week and time of day that particular bin should alert the user to take their pills.

Deprecated This is an ugly holdover from a previous system design. Move to using [DeviceBaseScheduleStrategy](#) directly. Refactoring in progress.

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/model/device/DeviceSchedule.java

4.37 jct.pillorganizer.repo.DeviceScheduleRepository Interface Reference

Public Member Functions

- List< [DeviceSchedule](#) > **findByDeviceId** (long id)
- List< [DeviceSchedule](#) > **findByDevice** ([Device](#) device)
- void **deleteByDevice** ([Device](#) device)
- int **update** (@Id [DeviceBinId](#) id, [DayOfWeek](#) dayOfWeek, int secondsFrom00, @Nullable [DeviceBaseDispenseTime](#) dispenseTime)

The documentation for this interface was generated from the following file:

- target/delombok/jct/pillorganizer/repo/DeviceScheduleRepository.java

4.38 jct.pillorganizer.service.DeviceScheduleService Class Reference

Public Member Functions

- SimpleScheduleDTO [buildSimpleSchedule](#) ([Device](#) device)
- SimpleScheduleDTO [updateSchedule](#) ([Device](#) device, SimpleScheduleDTO dto)
- LocalTime [fromSecondsFrom00](#) (long secondsFrom00)

4.38.1 Detailed Description

Business logic for dealing with device scheduling.

4.38.2 Member Function Documentation

4.38.2.1 buildSimpleSchedule()

```
SimpleScheduleDTO jct.pillorganizer.service.DeviceScheduleService.buildSimpleSchedule (
    Device device )
```

Fetches a device's simplified schedule DTO. If the device has no schedule registered, an empty schedule is returned.

Parameters

<i>device</i>	the device to fetch the schedule for
---------------	--------------------------------------

Returns

specified device's schedule in simplified DTO format

4.38.2.2 fromSecondsFrom00()

```
LocalTime jct.pillorganizer.service.DeviceScheduleService.fromSecondsFrom00 (
    long secondsFrom00 )
```

Converts a day-epoch format time into a LocalTime.

Parameters

<i>secondsFrom00</i>	day-epoch format to convert
----------------------	-----------------------------

Returns

converted LocalTime

4.38.2.3 updateSchedule()

```
SimpleScheduleDTO jct.pillorganizer.service.DeviceScheduleService.updateSchedule (
```

```
Device device,
SimpleScheduleDTO dto )
```

Updates a device's schedule based on the specified simplified schedule DTO. This function hides the complexity of the underlying schedule strategy class structure. If a device doesn't have a schedule strategy yet, a new one is created and persisted automatically.

Parameters

<i>device</i>	device to apply the schedule on
<i>dto</i>	a schedule DTO

Returns

a new schedule DTO that includes the new changes

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/service/DeviceScheduleService.java

4.39 jct.pillorganizer.repo.DeviceScheduleStrategyRepository Interface Reference

Public Member Functions

- Optional< [DeviceBaseScheduleStrategy](#) > **findByDevice** ([Device](#) device)

The documentation for this interface was generated from the following file:

- target/delombok/jct/pillorganizer/repo/DeviceScheduleStrategyRepository.java

4.40 jct.pillorganizer.service.DeviceService Class Reference

Public Member Functions

- [Device](#) **findOrCreateDevice** (long sn, [DeviceClass](#) deviceClass)

4.40.1 Detailed Description

Business logic for operations on Device objects.

4.40.2 Member Function Documentation

4.40.2.1 findOrCreateDevice()

```
Device jct.pillorganizer.service.DeviceService.findOrCreateDevice (
    long sn,
    DeviceClass deviceClass )
```

Fetches a Device object by serial number and class, if such device already exists, and if not, creates a new Device object with those specifications.

Parameters

<code>sn</code>	serial number of the device
<code>deviceClass</code>	class of the device

Returns

Device domain object, either existing or freshly persisted into the database

The documentation for this class was generated from the following file:

- `target/delombok/jct/pillorganizer/service/DeviceService.java`

4.41 `jct.pillorganizer.model.device.schedule.DeviceSimpleDispenseTime` Class Reference

Public Member Functions

- long `toSecondsFrom00` ()
- LocalTime `getTime` ()
- char `getPeriod` ()
- void `setTime` (final LocalTime time)
- void `setPeriod` (final char period)

Public Member Functions inherited from `jct.pillorganizer.model.device.schedule.DeviceBaseDispenseTime`

- Long `getId` ()
- `DeviceBaseScheduleStrategy` `getSchedule` ()
- List< `MedicationDispenseTime` > `getMedications` ()
- void `setId` (final Long id)
- void `setSchedule` (final `DeviceBaseScheduleStrategy` schedule)
- void `setMedications` (final List< `MedicationDispenseTime` > medications)

4.41.1 Detailed Description

A simple dispense time with a time and day period.

4.41.2 Member Function Documentation

4.41.2.1 `getPeriod()`

```
char jct.pillorganizer.model.device.schedule.DeviceSimpleDispenseTime.getPeriod ( )
```

The period of the day to schedule on. This determines whether this is scheduled on an AM bin or PM bin. Set to 'A' for AM and 'P' for PM.

4.41.2.2 `setPeriod()`

```
void jct.pillorganizer.model.device.schedule.DeviceSimpleDispenseTime.setPeriod (
    final char period )
```

The period of the day to schedule on. This determines whether this is scheduled on an AM bin or PM bin. Set to 'A' for AM and 'P' for PM.

4.41.2.3 `toSecondsFrom00()`

```
long jct.pillorganizer.model.device.schedule.DeviceSimpleDispenseTime.toSecondsFrom00 ( )
```

Converts the dispense time into day-epoch seconds.

Returns

the dispense time as the number of seconds from midnight, UTC

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/model/device/schedule/DeviceSimpleDispenseTime.java

4.42 `jct.pillorganizer.repo.DeviceSimpleDispenseTimeRepository` Interface Reference

Public Member Functions

- void **update** (@Id Long id, LocalTime time)

The documentation for this interface was generated from the following file:

- target/delombok/jct/pillorganizer/repo/DeviceSimpleDispenseTimeRepository.java

4.43 `jct.pillorganizer.model.device.schedule.DeviceSimpleSchedule`↵ Strategy Class Reference

Public Member Functions

- [DeviceSimpleDispenseTime](#) **amTime** ()
- [DeviceSimpleDispenseTime](#) **pmTime** ()
- [DeviceSimpleDispenseTime](#) **forPeriod** (char period)
- Object **buildDTO** ()

Public Member Functions inherited from

`jct.pillorganizer.model.device.schedule.DeviceBaseScheduleStrategy`

- abstract Object `buildDTO` ()
- Long `getId` ()
- `Device` `getDevice` ()
- Set< `DeviceBaseDispenseTime` > `getTimes` ()
- void `setId` (final Long id)
- void `setDevice` (final `Device` device)
- void `setTimes` (final Set< `DeviceBaseDispenseTime` > times)

4.43.1 Detailed Description

A simple medication schedule that is the same every day, with one time for AM and one time for PM.

4.43.2 Member Function Documentation

4.43.2.1 `amTime()`

```
DeviceSimpleDispenseTime jct.pillorganizer.model.device.schedule.DeviceSimpleScheduleStrategy.↔  
amTime ( )
```

Fetches the AM dispense time for this schedule.

Returns

AM dispense time, or null if none exists

4.43.2.2 `buildDTO()`

```
Object jct.pillorganizer.model.device.schedule.DeviceSimpleScheduleStrategy.buildDTO ( )
```

Serializes the schedule into a DTO object.

See also

`SimpleScheduleDTO`

Returns

a simplified representation of this schedule as a DTO object (POJO)

Reimplemented from `jct.pillorganizer.model.device.schedule.DeviceBaseScheduleStrategy`.

4.43.2.3 `forPeriod()`

```
DeviceSimpleDispenseTime jct.pillorganizer.model.device.schedule.DeviceSimpleScheduleStrategy.↔  
forPeriod (  
    char period )
```

Fetches the dispense time for a day period, if one exists.

Parameters

<i>period</i>	'A' for AM, 'P' for PM
---------------	------------------------

Returns

dispense time for the specified day period

4.43.2.4 pmTime()

```
DeviceSimpleDispenseTime jct.pillorganizer.model.device.schedule.DeviceSimpleScheduleStrategy.↔
pmTime ( )
```

Fetches the PM dispense time for this schedule.

Returns

PM dispense time, or null if none exists

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/model/device/schedule/DeviceSimpleScheduleStrategy.java

4.44 jct.pillorganizer.model.device.DeviceState Class Reference

Public Member Functions

- [DeviceBinId](#) **getId** ()
- [BinStatus](#) **getBinStatus** ()
- long **getScheduledTime** ()
- [Device](#) **getDevice** ()
- [DeviceEvent](#) **getEvent** ()
- [DeviceSchedule](#) **getSchedule** ()
- [DeviceBaseDispenseTime](#) **getDispenseTime** ()
- Long **getVersion** ()
- void **setId** (final [DeviceBinId](#) id)
- void **setBinStatus** (final [BinStatus](#) binStatus)
- void **setScheduledTime** (final long scheduledTime)
- void **setDevice** (final [Device](#) device)
- void **setEvent** (final [DeviceEvent](#) event)
- void **setSchedule** (final [DeviceSchedule](#) schedule)
- void **setDispenseTime** (final [DeviceBaseDispenseTime](#) dispenseTime)
- void **setVersion** (final Long version)

4.44.1 Detailed Description

Represents the state of a single "bin" on a particular pill organizer device. The key fields are the status, which indicates whether the bin is scheduled, taken, missed, etc, and the scheduled time, which indicates when that bin is scheduled to be opened. This structure is modelled after the bin state in the firmware.

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/model/device/DeviceState.java

4.45 jct.pillorganizer.device.DeviceStateJob Class Reference

Public Member Functions

- void **execute** ()

4.45.1 Detailed Description

Transitions device bin state statuses when appropriate. For example, it will move a pill from "PENDING" to "TAKE NOW" when appropriate. See backend documentation for details.

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/device/DeviceStateJob.java

4.46 jct.pillorganizer.repo.DeviceStateRepository Interface Reference

Public Member Functions

- List< [DeviceState](#) > **findByDevice** ([Device](#) device)
- List< [DeviceState](#) > **findByDeviceID** (long id)
- List< [DeviceState](#) > **findByDeviceIDWithSchedule** (long id)
- List< [DeviceState](#) > **findByDeviceAndTimeBetween** ([Device](#) d, long start, long end)
- void **deleteByDevice** ([Device](#) device)
- void **update** (@Id [DeviceBinId](#) id, @Version Long version, long scheduledTime, [BinStatus](#) binStatus)
- void **update** (@Id [DeviceBinId](#) id, @Version Long version, [BinStatus](#) binStatus)
- void **update** (@Id [DeviceBinId](#) id, @Version Long version, [BinStatus](#) binStatus, [DeviceEvent](#) event)
- void **update** (@Id [DeviceBinId](#) id, @Version Long version, [DeviceBaseDispenseTime](#) dispenseTime)
- void **updateResetState** (long device)
- List< [DeviceNotificationDetails](#) > **updatePendingToTakeNow** (long now)
- List< [DeviceNotificationDetails](#) > **updateTakeNowToMissed** (long threshold)
- List< [DeviceNotificationDetails](#) > **updateBinStateFromTime** (long setTo, long where, long threshold)

The documentation for this interface was generated from the following file:

- target/delombok/jct/pillorganizer/repo/DeviceStateRepository.java

4.47 jct.pillorganizer.service.DeviceStateService Class Reference

Public Member Functions

- [DeviceStateWrapper wrapperOf](#) ([Device](#) device)
- long [calculateStateFlags](#) ([Device](#) device)
- List< DosePeriodDTO > [buildDosePeriod](#) ([Device](#) device, [LocalDate](#) date)

4.47.1 Detailed Description

Business logic for dealing with device state.

4.47.2 Member Function Documentation

4.47.2.1 buildDosePeriod()

```
List< DosePeriodDTO > jct.pillorganizer.service.DeviceStateService.buildDosePeriod (
    Device device,
    LocalDate date )
```

Builds a compact representation of a device's state around a particular date. Only information relevant to that date is included in the resultant list.

Parameters

<i>device</i>	device to build the dose period for
<i>date</i>	the date that all dose periods should be on

Returns

a list of DosePeriodDTO objects, representing a time on the specified date that medication is taken

4.47.2.2 calculateStateFlags()

```
long jct.pillorganizer.service.DeviceStateService.calculateStateFlags (
    Device device )
```

Packs the status field of every bin state of a device into a single long. This is an optimization trip so that we don't need to store or serialize the status of every bin into an array of integers. Instead we only need one integer.

Parameters

<i>device</i>	device to calculate the state flag on
---------------	---------------------------------------

Returns

a long representing the packed status of every bin state

4.47.2.3 wrapperOf()

```
DeviceStateWrapper jct.pillorganizer.service.DeviceStateService.wrapperOf (
    Device device )
```

Creates a DeviceStateWrapper for a particular device.

Parameters

<i>device</i>	the device to create a wrapper around
---------------	---------------------------------------

Returns

a DeviceStateWrapper that operates on a particular device

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/service/DeviceStateService.java

4.48 jct.pillorganizer.device.DeviceStateWrapper Class Reference**Public Member Functions**

- **DeviceStateWrapper** ([DeviceRepository](#) deviceRepository, [DeviceScheduleRepository](#) scheduleRepository, [DeviceStateRepository](#) stateRepository, [DeviceEventRepository](#) deviceEventRepository, [FirmwareService](#) firmwareService, [Device](#) device)
- List< [DeviceState](#) > [getChronologicalState](#) ()
- Optional< [DeviceState](#) > [previousScheduled](#) ([DeviceState](#) state)
- Optional< [DeviceState](#) > [nextScheduled](#) ([DeviceState](#) state)
- void [handleBinEvent](#) ([DeviceEvent](#) event)
- void [updateState](#) (Pill.AllBinsState state)
- void [reload](#) ()
- Pill.SyncResponse [sync](#) (Pill.SyncRequest syncRequest)
- void [updateStateHash](#) ()
- Pill.AllBinsState.Builder [buildStateProtobuf](#) ()
- Iterable<? extends Pill.BinSchedule > [buildBinSchedule](#) ()
- void [initialize](#) (@Nullable Pill.AllBinsState initialState)
- void [buildInitialState](#) (@Nullable Pill.AllBinsState initialState)
- void [buildInitialSchedule](#) ()
- void [rebuildStateSchedule](#) ()
- List< [DeviceSchedule](#) > [getDeviceSchedule](#) ()
- List< [DeviceState](#) > [getState](#) ()

4.48.1 Detailed Description

Device state business logic.

4.48.2 Member Function Documentation

4.48.2.1 buildBinSchedule()

```
Iterable<? extends Pill.BinSchedule > jct.pillorganizer.device.DeviceStateWrapper.buildBinSchedule ( )
```

Serializes the device's schedule.

Returns

protobuf BinSchedule of the current device's dispense schedule

Deprecated use schedule strategy directly

4.48.2.2 buildInitialSchedule()

```
void jct.pillorganizer.device.DeviceStateWrapper.buildInitialSchedule ( )
```

Initializes the schedule of a new device, if a schedule doesn't exist. The freshly initialized schedule will have all bins disabled.

4.48.2.3 buildInitialState()

```
void jct.pillorganizer.device.DeviceStateWrapper.buildInitialState (
    @Nullable Pill.AllBinsState initialState )
```

Initialize the state of a new device. If a device already has a state, it will be preserved. The freshly initialized state will have all bins disabled.

Parameters

<i>initialState</i>	unused
---------------------	--------

4.48.2.4 buildStateProtobuf()

```
Pill.AllBinsState.Builder jct.pillorganizer.device.DeviceStateWrapper.buildStateProtobuf ( )
```

Serializes the device's state.

Returns

protobuf AllBinsState of the current device's state

4.48.2.5 getChronologicalState()

```
List< DeviceState > jct.pillorganizer.device.DeviceStateWrapper.getChronologicalState ( )
```

Get the device's bin states sorted by scheduled time, starting with the oldest

Returns

list of bin states in chronological order

4.48.2.6 getDeviceSchedule()

```
List< DeviceSchedule > jct.pillorganizer.device.DeviceStateWrapper.getDeviceSchedule ( )
```

Gets the device schedule.

Returns

all device bin schedules, in ascending order of bin ID

4.48.2.7 getState()

```
List< DeviceState > jct.pillorganizer.device.DeviceStateWrapper.getState ( )
```

Gets the device's bin states.

Returns

all device bin states, in ascending order of bin ID.

4.48.2.8 handleBinEvent()

```
void jct.pillorganizer.device.DeviceStateWrapper.handleBinEvent (
    DeviceEvent event )
```

Processes an event sourced from a device, with all state effects.

Parameters

<i>event</i>	event to process
--------------	------------------

4.48.2.9 initialize()

```
void jct.pillorganizer.device.DeviceStateWrapper.initialize (
    @Nullable Pill.AllBinsState initialState )
```

Initialize the state and schedule of a new device. If a device already had a state or schedule, it will be preserved. The initial state has all bins disabled and an empty schedule.

Parameters

<i>initialState</i>	optional state to initialize the device with
---------------------	--

4.48.2.10 nextScheduled()

```
Optional< DeviceState > jct.pillorganizer.device.DeviceStateWrapper.nextScheduled (
    DeviceState state )
```

Gets the next scheduled bin relative to another bin.

Parameters

<i>state</i>	bin to get the next scheduled bin for
--------------	---------------------------------------

Returns

the next scheduled bin, or empty if no next scheduled bin

4.48.2.11 previousScheduled()

```
Optional< DeviceState > jct.pillorganizer.device.DeviceStateWrapper.previousScheduled (
    DeviceState state )
```

Get the previously scheduled bin relative to another bin.

Parameters

<i>state</i>	bin to get the previous scheduled bin for
--------------	---

Returns

the previously scheduled bin, or empty if no previously scheduled bin

4.48.2.12 rebuildStateSchedule()

```
void jct.pillorganizer.device.DeviceStateWrapper.rebuildStateSchedule ( )
```

Synchronizes the device's state scheduled times with the device's schedule. Used, for example, if the schedule is changed in the middle of the week.

4.48.2.13 reload()

```
void jct.pillorganizer.device.DeviceStateWrapper.reload ( )
```

Reload the pills in a device by resetting the state's schedule.

4.48.2.14 sync()

```
Pill.SyncResponse jct.pillorganizer.device.DeviceStateWrapper.sync (
    Pill.SyncRequest syncRequest )
```

Performs a two-way device sync according to the business rules.

Parameters

<i>syncRequest</i>	protobuf sync request structure containing the device's state and events to process
--------------------	---

Returns

sync response structure the device should process and use to override its internal state with

4.48.2.15 updateState()

```
void jct.pillorganizer.device.DeviceStateWrapper.updateState (
    Pill.AllBinsState state )
```

Override the bin state stored in the database with a Protobuf-encoded state structure.

Parameters

<i>state</i>	the state to override server state with
--------------	---

4.48.2.16 updateStateHash()

```
void jct.pillorganizer.device.DeviceStateWrapper.updateStateHash ( )
```

Update the state hash in the database to match the current state stored in this object.

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/device/DeviceStateWrapper.java

4.49 jct.pillorganizer.model.device.DeviceUser Class Reference

Public Member Functions

- Long **getId** ()
- [Device](#) **getDevice** ()
- [BaseUser](#) **getUser** ()
- long **getUserID** ()
- long **getDeviceID** ()
- boolean **isPrimaryUser** ()
- boolean **isOwner** ()

- String **getNotificationToken** ()
- void **setId** (final Long id)
- void **setDevice** (final [Device](#) device)
- void **setUser** (final [BaseUser](#) user)
- void **setUserID** (final long userID)
- void **setDeviceID** (final long deviceID)
- void **setPrimaryUser** (final boolean primaryUser)
- void **setOwner** (final boolean owner)
- void **setNotificationToken** (final String notificationToken)

4.49.1 Detailed Description

Relates a [BaseUser](#) to a [Device](#). The role the user has to a device is recorded here, whether they are the designated primary user of the device, etc. The notification token for push notification delivery is also stored here. Thus, push notification delivery is routed on a per-device per-user basis. TODO: move notification token to another entity to resolve notification counter-intuitiveness

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/model/device/DeviceUser.java

4.50 jct.pillorganizer.repo.DeviceUserAsyncRepository Interface Reference

Public Member Functions

- Mono< [DeviceUser](#) > **findByUserIDAndDeviceID** (long user, long device)
- Mono< [Device](#) > **retrieveDeviceByUserIDAndDeviceID** (long userID, long deviceID)

The documentation for this interface was generated from the following file:

- target/delombok/jct/pillorganizer/repo/DeviceUserAsyncRepository.java

4.51 jct.pillorganizer.repo.DeviceUserRepository Interface Reference

Public Member Functions

- int **countByUserIDAndDeviceID** (long user, long device)
- int **countByUserAndDevice** ([BaseUser](#) user, [Device](#) device)
- void **update** (@Id Long id, String notificationToken)
- void **updateNotificationTokenById** (@Id Long id, @Nullable String notificationToken)
- [DeviceUser](#) **findByUserIDAndDeviceID** (long user, long device)
- Optional< [Device](#) > **retrieveDeviceByUserIDAndDeviceID** (long userID, long deviceID)
- Set< DeviceUserDTO > **findByUserID** (long user)
- Optional< DeviceUserDTO > **retrieveByUserIDAndDeviceID** (long user, long device)

The documentation for this interface was generated from the following file:

- target/delombok/jct/pillorganizer/repo/DeviceUserRepository.java

4.52 jct.pillorganizer.service.DeviceUserService Class Reference

Public Member Functions

- void [addUserToDevice](#) (long userID, long deviceID, boolean owner, boolean primaryUser)
- boolean [doesUserBelongToDevice](#) ([BaseUser](#) user, [Device](#) device)
- boolean [doesUserBelongToDevice](#) (long userID, long deviceID)

4.52.1 Detailed Description

Business logic for `DeviceUser` relationships.

4.52.2 Member Function Documentation

4.52.2.1 `addUserToDevice()`

```
void jct.pillorganizer.service.DeviceUserService.addUserToDevice (
    long userID,
    long deviceID,
    boolean owner,
    boolean primaryUser )
```

Adds a user to a device.

Parameters

<i>userID</i>	user ID to add to the device
<i>deviceID</i>	device ID to add the user to
<i>owner</i>	whether the user has "owner" privileges over the device
<i>primaryUser</i>	whether the user should be marked as the primary user of the device

4.52.2.2 `doesUserBelongToDevice()` [1/2]

```
boolean jct.pillorganizer.service.DeviceUserService.doesUserBelongToDevice (
    BaseUser user,
    Device device )
```

Checks if a user is added to a specific device.

Parameters

<i>user</i>	the user to test membership
<i>device</i>	the device to see whether the user is related to it or not

Returns

true if the user is related to the device

4.52.2.3 doesUserBelongToDevice() [2/2]

```
boolean jct.pillorganizer.service.DeviceUserService.doesUserBelongToDevice (
    long userID,
    long deviceID )
```

Checks if a user is added to a specific device.

Parameters

<i>userID</i>	the user ID to test membership
<i>deviceID</i>	the device ID to see whether the user is related to it or not

Returns

true if the user is related to the device

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/service/DeviceUserService.java

4.53 jct.pillorganizer.dto.DeviceUserSettings Class Reference

Public Member Functions

- void **setNotificationToken** (String title)
- boolean **setNotificationToken** ()

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/dto/DeviceUserSettings.java

4.54 jct.pillorganizer.model.EventType Enum Reference

Public Member Functions

- **EventType** (int val)
- int [getIntValue](#) ()

Static Public Member Functions

- static [EventType fromProtobuf](#) (Pill.RecordedEvent.EventType eventType)

Public Attributes

- **OPENED** =(0)
- **CLOSED** =(1)
- **MISSED** =(2)
- **RELOAD** =(3)

4.54.1 Detailed Description

Enumeration of different event types received from pill organizers for use in our database models.

Deprecated should use protobuf directly

4.54.2 Member Function Documentation

4.54.2.1 fromProtobuf()

```
static EventType jct.pillorganizer.model.EventType.fromProtobuf (
    Pill.RecordedEvent.EventType eventType ) [static]
```

Convert from an EventType as described in the protocol buffers to this enumeration

Parameters

<i>eventType</i>	a protobuf EventType
------------------	----------------------

Returns

a Java EventType

4.54.2.2 getIntValue()

```
int jct.pillorganizer.model.EventType.getIntValue ( )
```

Converts the event type into the firmware event type ID

Returns

firmware event type ID for this event type

Deprecated

The documentation for this enum was generated from the following file:

- target/delombok/jct/pillorganizer/model/EventType.java

4.55 jct.pillorganizer.service.FirebaseMessageService Class Reference

Public Member Functions

- **FirebaseMessageService** (@Value("\${firebase.privkey}") String privKey, @Value("\${firebase.project}") String projID, @Value("\${firebase.email}") String email) throws IOException
- void [sendPillReminderNotification](#) (DeviceNotificationDetails details)

4.55.1 Detailed Description

Service to send push notifications to registered mobile phones.

4.55.2 Member Function Documentation

4.55.2.1 [sendPillReminderNotification\(\)](#)

```
void jct.pillorganizer.service.FirebaseMessageService.sendPillReminderNotification (
    DeviceNotificationDetails details )
```

Sends a pill reminder notification to a specific notification token.

Parameters

<i>details</i>	a structure containing the info necessary to deliver a push notification
----------------	--

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/service/FirebaseMessageService.java

4.56 jct.pillorganizer.service.FirmwareService Class Reference

Public Member Functions

- int [getLatestVersion](#) ()

4.56.1 Detailed Description

Firmware service

4.56.2 Member Function Documentation

4.56.2.1 `getLatestVersion()`

```
int jct.pillorganizer.service.FirmwareService.getLatestVersion ( )
```

Returns the latest version of the CabiNET firmware.

Returns

latest version of the CabiNET firmware, as an integer.

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/service/FirmwareService.java

4.57 `jct.pillorganizer.serde.HexEncodeSerde` Class Reference

Public Member Functions

- `byte[] deserialize` (Decoder decoder, DecoderContext context, Argument<? super byte[]> type) throws IOException
- `void serialize` (Encoder encoder, EncoderContext context, Argument<? extends byte[]> type, byte[] value) throws IOException

4.57.1 Detailed Description

Serializes a byte array into a hex string.

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/serde/HexEncodeSerde.java

4.58 `jct.pillorganizer.auth.JwkController` Class Reference

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/auth/JwkController.java

4.59 `jct.pillorganizer.auth.JwtIssuerService` Class Reference

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/auth/JwtIssuerService.java

4.60 jct.pillorganizer.serde.LocalTimeEncodeSerde Class Reference

Public Member Functions

- LocalTime **deserialize** (Decoder decoder, DecoderContext context, Argument<? super LocalTime > type) throws IOException
- void **serialize** (Encoder encoder, EncoderContext context, Argument<? extends LocalTime > type, LocalTime value) throws IOException

4.60.1 Detailed Description

Serializes a LocalTime into day-epoch format.

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/serde/LocalTimeEncodeSerde.java

4.61 jct.pillorganizer.auth.LoginController Class Reference

Public Member Functions

- Publisher< MutableHttpResponse<?> > **loginAnonymous** (@Valid @Body [AnonymousAuthenticationRequest](#) creds, HttpRequest<?> request)
- Publisher< MutableHttpResponse<?> > **login** (@Valid @Body [UserPassAuthenticationRequest](#) creds, HttpRequest<?> request)
- Publisher< MutableHttpResponse<?> > **login** (@Body byte[] body, HttpRequest<?> request)

4.61.1 Detailed Description

Handles logins via HTTP.

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/auth/LoginController.java

4.62 jct.pillorganizer.model.medication.MedicationDispenseTime Class Reference

Public Member Functions

- Long **getId** ()
- [ScheduledMedication](#) **getMedication** ()
- [DeviceBaseDispenseTime](#) **getDispense** ()
- long **getMedicationID** ()
- long **getDispenseID** ()
- int **getQuantity** ()
- void **setId** (final Long id)
- void **setMedication** (final [ScheduledMedication](#) medication)
- void **setDispense** (final [DeviceBaseDispenseTime](#) dispense)
- void **setMedicationID** (final long medicationID)
- void **setDispenseID** (final long dispenseID)
- void **setQuantity** (final int quantity)

4.62.1 Detailed Description

Relates a `ScheduledMedication` to a `DeviceBaseDispenseTime`. Also contains the quantity of medication to be taken at a given dispense time.

See also

`ScheduledMedication`
`DeviceBaseDispenseTime`

The documentation for this class was generated from the following file:

- `target/delombok/jct/pillorganizer/model/medication/MedicationDispenseTime.java`

4.63 jct.pillorganizer.repo.MedicationDispenseTimeRepository Interface Reference

Public Member Functions

- Optional< [MedicationDispenseTime](#) > **findByDispenseIDAndMedicationID** (long dispenseID, long medicationID)
- List< [MedicationDispenseTime](#) > **findByMedicationID** (long medicationID)
- void **deleteById** (long id)

The documentation for this interface was generated from the following file:

- `target/delombok/jct/pillorganizer/repo/MedicationDispenseTimeRepository.java`

4.64 jct.pillorganizer.service.MedicationService Class Reference

Public Member Functions

- void **delete** (long medicationID, [Device](#) device)
- [ScheduledMedication](#) **saveFromDto** ([Device](#) d, SaveMedicationDTO dto)

4.64.1 Detailed Description

Business logic for operations on `ScheduledMedication` objects.

4.64.2 Member Function Documentation

4.64.2.1 delete()

```
void jct.pillorganizer.service.MedicationService.delete (
    long medicationID,
    Device device )
```

Deletes a single `ScheduledMedication` (and their dependents) by ID.

Parameters

<i>medicationID</i>	the ID of the medication to delete
<i>device</i>	the device ID of the device the medication is associated with.

4.64.2.2 saveFromDto()

```
ScheduledMedication jct.pillorganizer.service.MedicationService.saveFromDto (
    Device d,
    SaveMedicationDTO dto )
```

Creates or updates a medication associated with a device. If a medication exists with the ID specified in the DTO object, it will be updated with the new information in the DTO. Otherwise, a new medication is created.

Parameters

<i>d</i>	device the medication is associated with
<i>dto</i>	DTO object of the medication to persist

Returns

domain entity of the persisted medication

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/service/MedicationService.java

4.65 jct.pillorganizer.model.medication.MedicationShape Enum Reference

Public Attributes

- CAPSULE
- DIAMOND
- DOUBLE_CIRCLE
- HEPTAGON
- HEXAGON
- OCTAGON
- OVAL
- PENTAGON
- RECTANGLE
- ROUND
- SEMICIRCLE
- SQUARE
- TEAR
- TRAPEZOID
- TRIANGLE

The documentation for this enum was generated from the following file:

- target/delombok/jct/pillorganizer/model/medication/MedicationShape.java

4.66 jct.pillorganizer.dto.NotificationToken Class Reference

Public Member Functions

- **NotificationToken** (final String notificationToken)
- String **getNotificationToken** ()
- boolean **equals** (final java.lang.Object o)
- int **hashCode** ()
- java.lang.String **toString** ()

Public Attributes

- final String **notificationToken**

Protected Member Functions

- boolean **canEqual** (final java.lang.Object other)

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/dto/NotificationToken.java

4.67 jct.pillorganizer.dto.ProvisionStatus Class Reference

Public Member Functions

- **ProvisionStatus** (final boolean provisioned, final long deviceId)
- boolean **isProvisioned** ()
- long **getDeviceID** ()
- boolean **equals** (final java.lang.Object o)
- int **hashCode** ()
- java.lang.String **toString** ()

Protected Member Functions

- boolean **canEqual** (final java.lang.Object other)

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/dto/ProvisionStatus.java

4.68 jct.pillorganizer.model.medication.ScheduledMedication Class Reference

Public Member Functions

- Long **getId** ()
- [Device](#) **getDevice** ()
- Set< [MedicationDispenseTime](#) > **getDispenseTimes** ()
- long **getDeviceID** ()
- String **getMed_name** ()
- [MedicationShape](#) **getShape** ()
- Integer **getColor** ()
- void **setId** (final Long id)
- void **setDevice** (final [Device](#) device)
- void **setDispenseTimes** (final Set< [MedicationDispenseTime](#) > dispenseTimes)
- void **setDeviceID** (final long deviceID)
- void **setMed_name** (final String med_name)
- void **setShape** (final [MedicationShape](#) shape)
- void **setColor** (final Integer color)

4.68.1 Detailed Description

A medication that is taken according to some schedule. We currently view a "medication" to be anything you could put in a pill organizer - currently we only really consider pills under this view. A `ScheduledMedication` has a name, shape, color, and a frequency and periodicity.

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/model/medication/ScheduledMedication.java

4.69 jct.pillorganizer.repo.ScheduledMedicationAsyncRepository Interface Reference

Public Member Functions

- Flux< [ScheduledMedication](#) > **findByDevice** ([Device](#) device)
- Flux< [ScheduledMedication](#) > **retrieveByDevice** ([Device](#) device)
- Mono< [ScheduledMedication](#) > **retrieveByDeviceAndId** ([Device](#) device, long id)
- Mono< Long > **deleteByIdAndDevice** (long id, [Device](#) device)

The documentation for this interface was generated from the following file:

- target/delombok/jct/pillorganizer/repo/ScheduledMedicationAsyncRepository.java

4.70 jct.pillorganizer.repo.ScheduledMedicationRepository Interface Reference

Public Member Functions

- [ScheduledMedication](#) **retrieveById** (long id)
- void **update** (@Id Long id, String med_name, @Nullable [MedicationShape](#) shape, @Nullable Integer color)
- List< [ScheduledMedication](#) > **retrieveByDevice** ([Device](#) device)
- [ScheduledMedication](#) **retrieveByDeviceAndId** ([Device](#) device, long id)
- void **deleteByIdAndDevice** (long id, [Device](#) device)
- Optional< [ScheduledMedication](#) > **findByIdAndDevice** (long id, [Device](#) device)

The documentation for this interface was generated from the following file:

- target/delombok/jct/pillorganizer/repo/ScheduledMedicationRepository.java

4.71 jct.pillorganizer.serde.SerialNumberSerde Class Reference

Public Member Functions

- Long **deserialize** (Decoder decoder, DecoderContext context, Argument<? super Long > type) throws IOException
- void **serialize** (Encoder encoder, EncoderContext context, Argument<? extends Long > type, Long value) throws IOException

4.71.1 Detailed Description

Serializes a serial number (internally stored as a long) into a hex string. A serial number is 6 bytes and a long is 8 bytes, so the long is truncated.

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/serde/SerialNumberSerde.java

4.72 jct.pillorganizer.dto.StartProvision Class Reference

Public Member Functions

- long **getSerialNo** ()
- String **getDeviceClass** ()
- void **setSerialNo** (final long serialNo)
- void **setDeviceClass** (final String deviceClass)
- boolean **equals** (final java.lang.Object o)
- int **hashCode** ()
- java.lang.String **toString** ()

Protected Member Functions

- boolean **canEqual** (final java.lang.Object other)

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/dto/StartProvision.java

4.73 jct.pillorganizer.model.user.User Class Reference

Public Member Functions

- String **getEmail** ()
- byte[] **getPasswordHash** ()
- void **setEmail** (final String email)
- void **setPasswordHash** (final byte[] passwordHash)

Public Member Functions inherited from [jct.pillorganizer.model.user.BaseUser](#)

- Long **getId** ()
- List< [DeviceUser](#) > **getDevices** ()
- [UserRole](#) **getRole** ()
- Timestamp **getCreated** ()
- Timestamp **getUpdated** ()
- void **setId** (final Long id)
- void **setDevices** (final List< [DeviceUser](#) > devices)
- void **setRole** (final [UserRole](#) role)
- void **setCreated** (final Timestamp created)
- void **setUpdated** (final Timestamp updated)

4.73.1 Detailed Description

A standard user account with an email/password combo. Users of this type are intended to be humans who explicitly registered for a CabiNET account and voluntarily supplied their own email and password.

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/model/user/User.java

4.74 jct.pillorganizer.auth.UserPassAuthenticationProvider Class Reference

Public Member Functions

- Publisher< AuthenticationResponse > **authenticate** (HttpRequest<?> httpReq, AuthenticationRequest<?, ?> req)

4.74.1 Detailed Description

Implements anonymous user authentication using emails and passwords. The username is the user's email. Roles are granted based on the user's role field in their database record and are assigned the "id" attribute for their user ID.

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/auth/UserPassAuthenticationProvider.java

4.75 jct.pillorganizer.auth.UserPassAuthenticationRequest Class Reference

4.75.1 Detailed Description

Standard user authentication request to differentiate from the other authentication types.

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/auth/UserPassAuthenticationRequest.java

4.76 jct.pillorganizer.dto.UserRegistration Class Reference

Public Member Functions

- String **getEmail** ()
- String **getPassword** ()
- void **setEmail** (final String email)
- void **setPassword** (final String password)
- boolean **equals** (final java.lang.Object o)
- int **hashCode** ()
- java.lang.String **toString** ()

Protected Member Functions

- boolean **canEqual** (final java.lang.Object other)

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/dto/UserRegistration.java

4.77 jct.pillorganizer.repo.UserRepository Interface Reference

Public Member Functions

- Mono< [User](#) > **findByEmail** (String email)
- Mono< Integer > **countByEmail** (String email)
- Mono< Integer > **upgradeAnonymousUser** (long id, String email, byte[] passwordHash)
- Mono< UserInfoDTO > **findUserInfoDTOFromID** (long id)

The documentation for this interface was generated from the following file:

- target/delombok/jct/pillorganizer/repo/UserRepository.java

4.78 jct.pillorganizer.model.user.UserRole Enum Reference

Public Attributes

- **USER**
- **ADMIN**

4.78.1 Detailed Description

An enumeration of possible special roles for a user.

The documentation for this enum was generated from the following file:

- target/delombok/jct/pillorganizer/model/user/UserRole.java

4.79 jct.pillorganizer.model.user.UserType Enum Reference

Public Attributes

- **STANDARD**
- **DEVICE**
- **ANONYMOUS**

4.79.1 Detailed Description

The purpose of this enumeration was to differentiate between devices (which can login) and regular users. This is a bad idea and must be refactored out.

Deprecated

The documentation for this enum was generated from the following file:

- target/delombok/jct/pillorganizer/model/user/UserType.java

4.80 jct.pillorganizer.dto.VerifyProvision Class Reference

Public Member Functions

- long **getSerialNo** ()
- String **getSsid** ()
- void **setSerialNo** (final long serialNo)
- void **setSsid** (final String ssid)
- boolean **equals** (final java.lang.Object o)
- int **hashCode** ()
- java.lang.String **toString** ()

Protected Member Functions

- boolean **canEqual** (final java.lang.Object other)

The documentation for this class was generated from the following file:

- target/delombok/jct/pillorganizer/dto/VerifyProvision.java

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